RFC: SAP

Summary

This RFC proposes a context-aware guardrail architecture for AI-enhanced SAP workflows, in which every request is routed through a controlled gateway based on the SAP module, data sensitivity, and the type of action being taken. Since SAP spans highly sensitive and tightly connected areas like finance, procurement, sales, operations, and HR, AI access can be incredibly valuable but also carries real risk.

Finance Schema

```
finance_records (
  id uuid primary key,
  sap_record_id text,
  amount numeric,
  currency_code text,
  status text,
  sensitivity_class text,
  created_at timestamptz,
```

```
  updated_at timestamptz
)
```

Procurement Schema

```
procurement_records (
  id uuid primary key,
  sap_record_id text,
  supplier_id text,
  quantity numeric,
  status text,
  sensitivity_class text,
  created_at timestamptz,
  updated_at timestamptz
)
```

Sales Schema

```
sales_records (
  id uuid primary key,
  sap_record_id text,
  customer_id text,
  net_amount numeric,
  status text,
  sensitivity_class text,
  created_at timestamptz,
  updated_at timestamptz
)
```

HR Schema

```
hr_records (
  id uuid primary key,
```

```
  sap_record_id text,
  employee_id text,
  employment_status text,
  status text,
  sensitivity_class text,
  created_at timestamptz,
  updated_at timestamptz
)
```

Operations Schema

```
operations_records (
  id uuid primary key,
  sap_record_id text,
  material_id text,
  status text,
  sensitivity_class text,
  created_at timestamptz,
  updated_at timestamptz
)
```

Status Schema

```
pending
synced
error
```

Sensitivity Class Schema

```
low
high
```

Proposed Architecture

Supabase

Stores workflow state, action requests, outputs, and UI facing status. It is the main persistence layer for application records and AI workflow state.

Edge function trigger

Triggered when a qualifying table update occurs. Its job is not to make policy decisions itself, but to respond to state changes and initiate the downstream AI workflow.

Internal API gateway

Core of the architecture. It sits between app events and model routing.

Lava request layer

Lava is used as the external model gateway. It serves as the deterministic routing plane through which requests are sent to approved model providers.

Goal of the UI layer

The UI should allow a user to trigger business actions in a controlled, observable way, while rendering AI generated output back into the same operational surface.

UI record model

```
action_requests (
  id uuid primary key,
  source_record_id uuid,
  action_type text,
  status text,
  created_at timestamptz
  updated_at timestamptz
)
```

```
action_type text check (
  action_type in (
    'summarize',
    'draft',
    'send',
    'review'
  )
)
```

Example: sending an order

1. User opens an order page.
2. User clicks **Send order**.
3. Frontend writes an action request row to Supabase.
4. Edge function detects the new request.
5. The internal gateway classifies the request as sales-domain and chooses the proper route.
6. Lava forwards the request to the approved model path.
7. The returned draft or output is stored in Supabase.
8. The UI refreshes or subscribes to the output.
9. User reviews the generated content.
10. User approves the final send action.